

AI 2002: Artificial Intelligence

Machine Learning

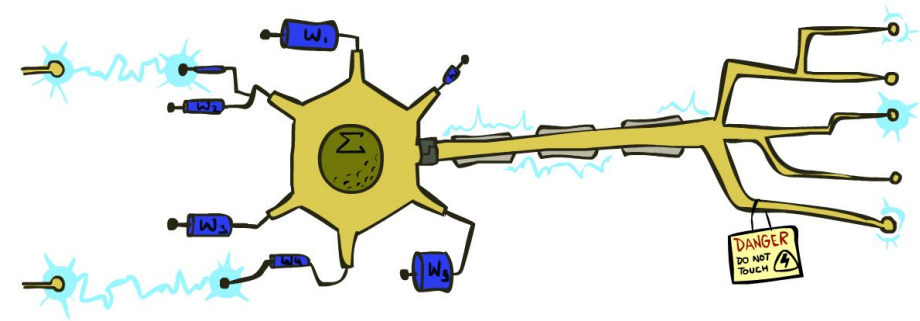
Spring 2026

[These slides (mostly) are based on those created by Dan Klein and Pieter Abbeel (ai.berkeley.edu).]

[Some slides are by Mr. Sandesh Kumar, and Ms. Anaum Hamid, FAST-NUCES, Karahi, and by Nitin Agarwal, Software Consultant]

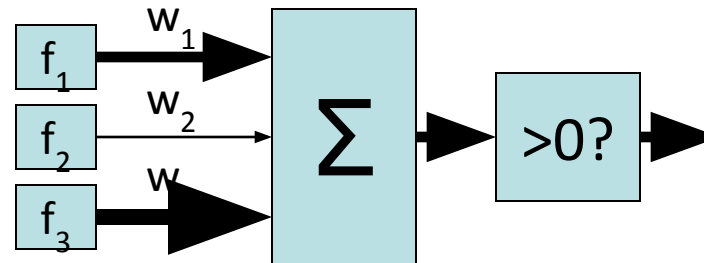
Reminder: Linear Classifiers

- Inputs are **feature values**
- Each feature has a **weight**
- Sum is the **activation**



$$\text{activation}_w(x) = \sum_i w_i \cdot f_i(x) = w \cdot f(x)$$

- If the activation is:
 - Positive, output +1
 - Negative, output -1

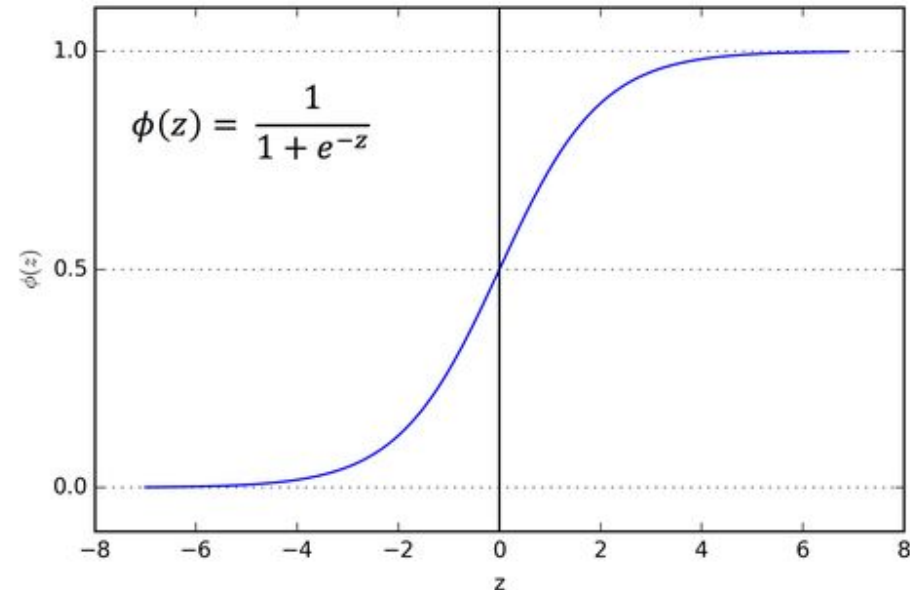


How to get probabilistic decisions?

- Activation: $z = w \cdot f(x)$
- If $z = w \cdot f(x)$ very positive ☐ want probability going to 1
- If $z = w \cdot f(x)$ very negative ☐ want probability going to 0

- Sigmoid function

$$\phi(z) = \frac{1}{1 + e^{-z}}$$



Best w ?

- Maximum likelihood estimation:

$$\max_w ll(w) = \max_w \sum_i \log P(y^{(i)} | x^{(i)}; w)$$

with:

$$P(y^{(i)} = +1 | x^{(i)}; w) = \frac{1}{1 + e^{-w \cdot f(x^{(i)})}}$$
$$P(y^{(i)} = -1 | x^{(i)}; w) = 1 - \frac{1}{1 + e^{-w \cdot f(x^{(i)})}}$$

= Logistic Regression

- Optimization

- i.e., how do we solve:

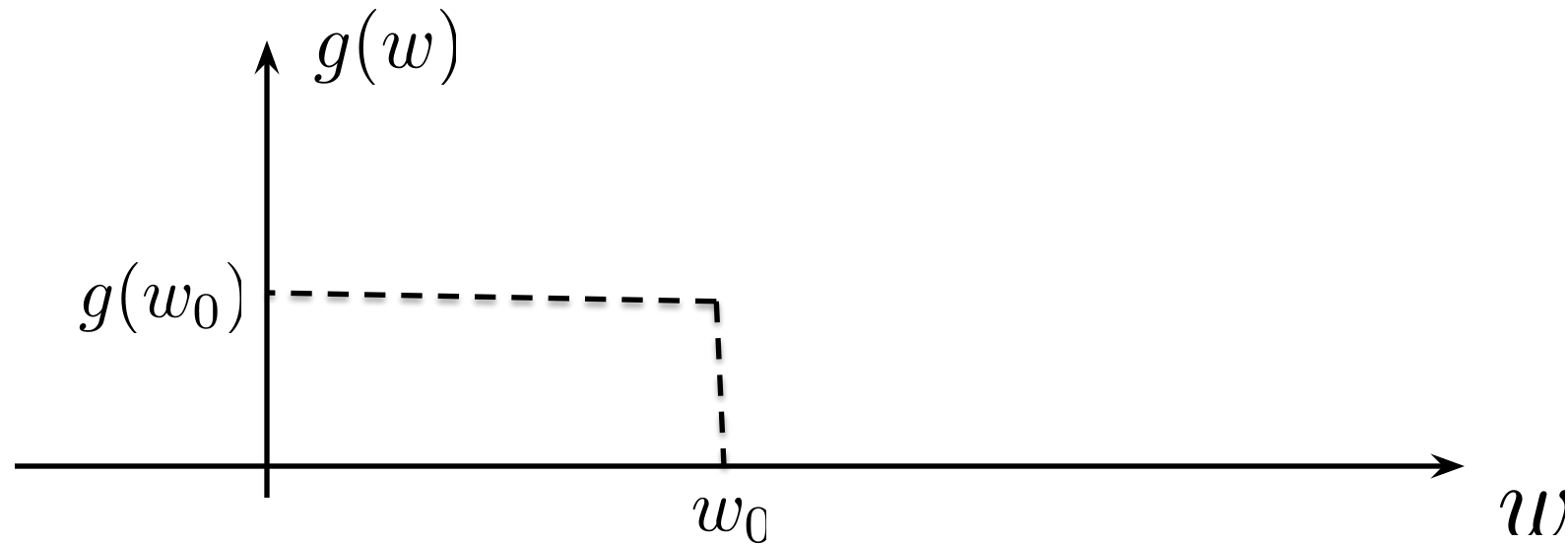
$$\max_w ll(w) = \max_w \sum_i \log P(y^{(i)} | x^{(i)}; w)$$

Hill Climbing

- Recall from CSPs lecture: simple, general
 - Start wherever
 - Repeat: move to the best neighboring state
 - If no neighbors better than current, quit
- What's particularly tricky when hill-climbing for multiclass logistic regression?
 - Optimization over a continuous space
 - Infinitely many neighbors!
 - How to do this efficiently?

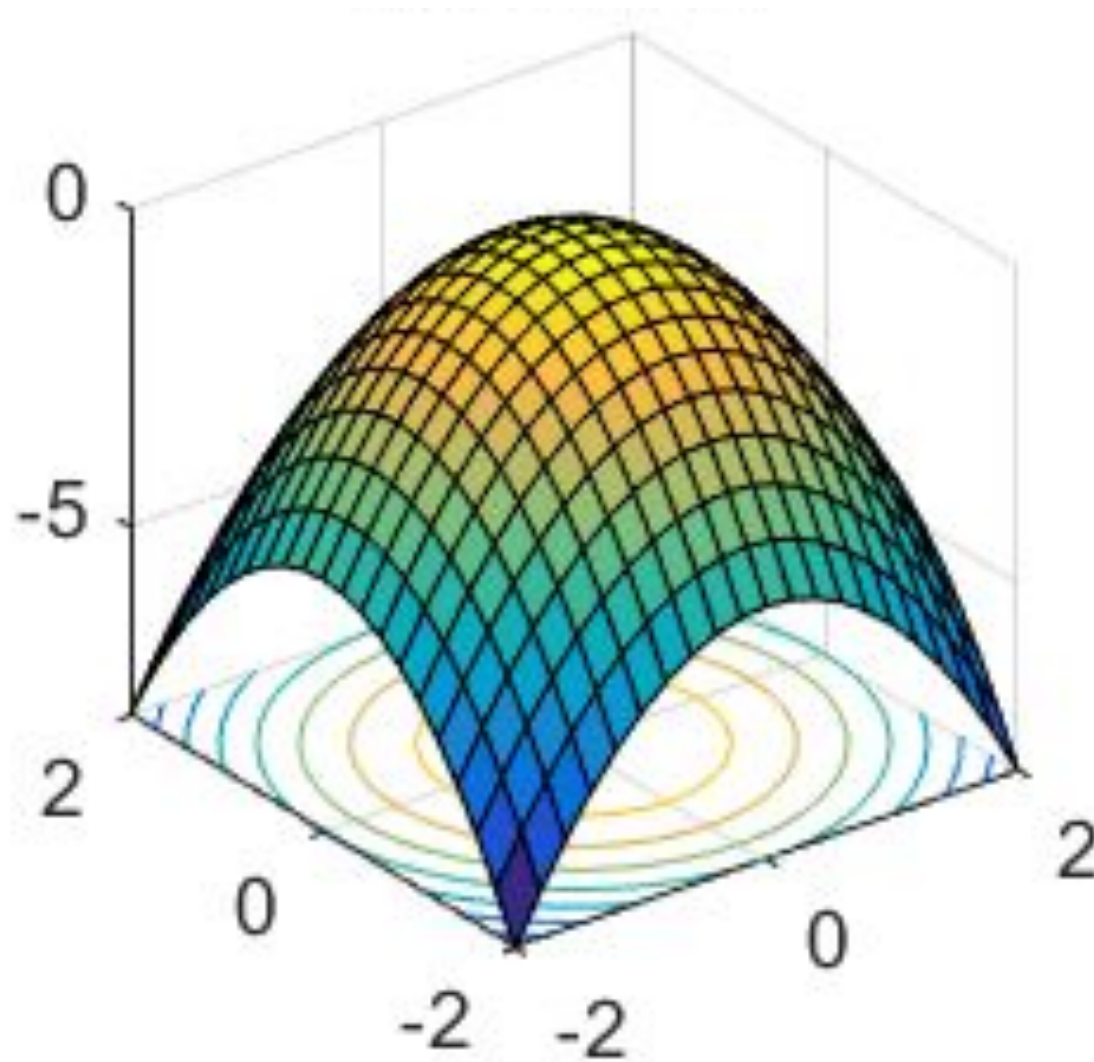


1-D Optimization



- Could evaluate $g(w_0 + h)$ and $g(w_0 - h)$
 - Then step in best direction
- Or, evaluate derivative:
$$\frac{\partial g(w_0)}{\partial w} = \lim_{h \rightarrow 0} \frac{g(w_0 + h) - g(w_0 - h)}{2h}$$
 - Tells which direction to step into

2-D Optimization



Gradient Ascent

- Perform update in uphill direction for each coordinate
- The steeper the slope (i.e. the higher the derivative) the bigger the step for that coordinate
- E.g., consider the function $g(w_1, w_2)$

- Updates:

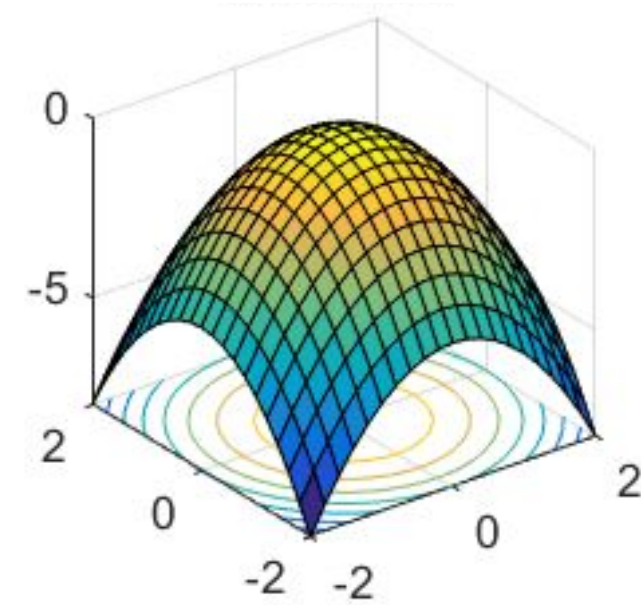
$$w_1 \leftarrow w_1 + \alpha * \frac{\partial g}{\partial w_1}(w_1, w_2)$$

$$w_2 \leftarrow w_2 + \alpha * \frac{\partial g}{\partial w_2}(w_1, w_2)$$

- Updates in vector notation:

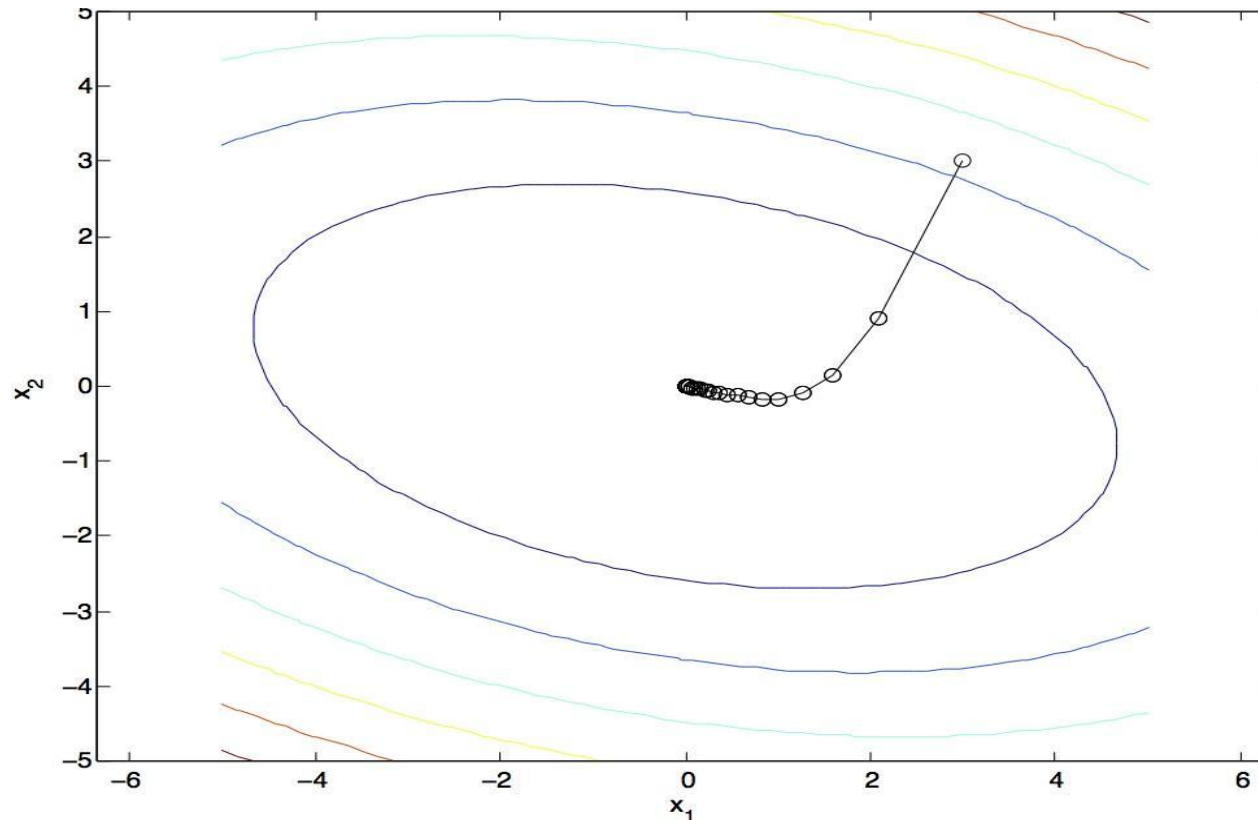
$$w \leftarrow w + \alpha * \nabla_w g(w)$$

with: $\nabla_w g(w) = \begin{bmatrix} \frac{\partial g}{\partial w_1}(w) \\ \frac{\partial g}{\partial w_2}(w) \end{bmatrix}$ = gradient

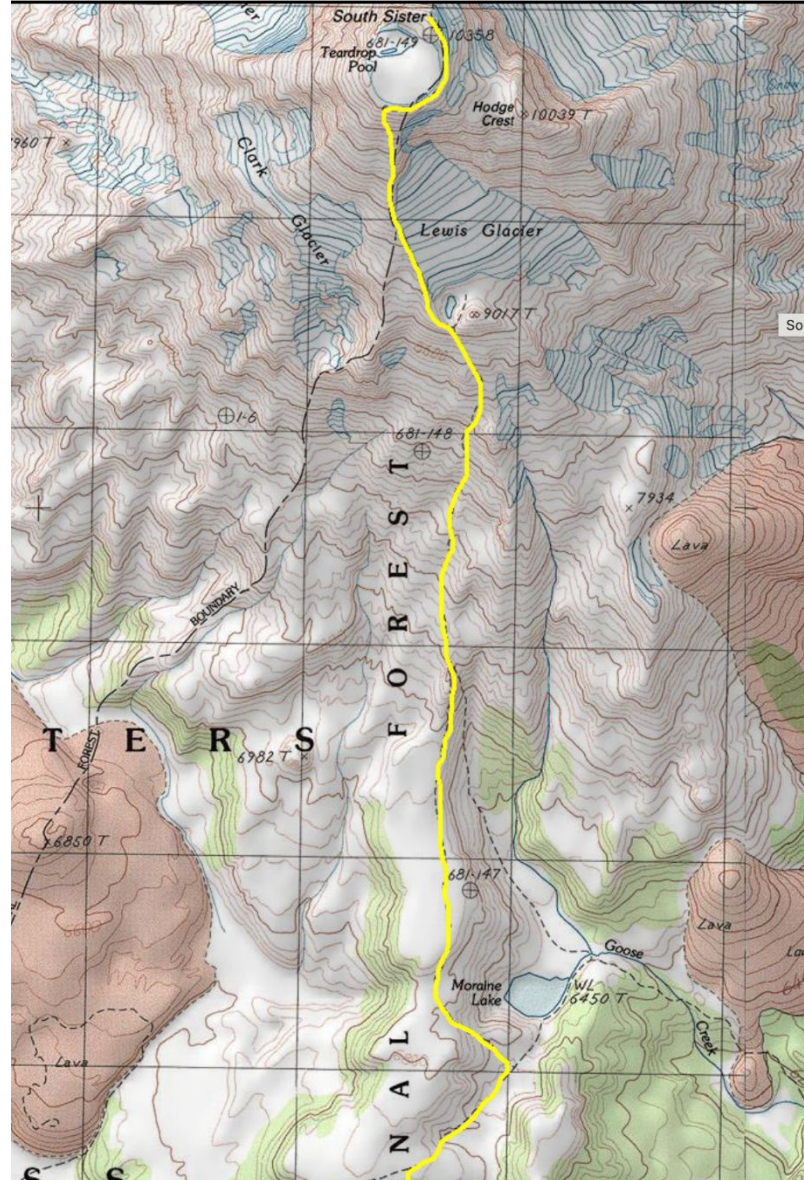


Gradient Ascent

- Idea:
 - Start somewhere
 - Repeat: Take a step in the gradient direction



What is the Steepest Direction? Hiker's intuition



What is the Steepest Direction? With actual math



$$\max_{\Delta: \Delta_1^2 + \Delta_2^2 \leq \varepsilon} g(w + \Delta)$$

- First-Order Taylor Expansion:

$$g(w + \Delta) \approx g(w) + \frac{\partial g}{\partial w_1} \Delta_1 + \frac{\partial g}{\partial w_2} \Delta_2$$

- Steepest Descent Direction:

$$\max_{\Delta: \Delta_1^2 + \Delta_2^2 \leq \varepsilon} g(w) + \frac{\partial g}{\partial w_1} \Delta_1 + \frac{\partial g}{\partial w_2} \Delta_2$$

- Recall: $\max_{\Delta: \|\Delta\| \leq \varepsilon} \Delta^\top a$

$$\Delta = \varepsilon \frac{a}{\|a\|}$$

- Hence, solution: $\Delta = \varepsilon \frac{\nabla g}{\|\nabla g\|}$

Gradient direction = steepest direction!

$$\nabla g = \begin{bmatrix} \frac{\partial g}{\partial w_1} \\ \frac{\partial g}{\partial w_2} \end{bmatrix}$$

Gradient in n dimensions

$$\nabla g = \begin{bmatrix} \frac{\partial g}{\partial w_1} \\ \frac{\partial g}{\partial w_2} \\ \dots \\ \frac{\partial g}{\partial w_n} \end{bmatrix}$$

Example: Derivatives and Gradients

- What is the derivative of the function $g(x) = x^2 + 3$

$$\frac{dg}{dx} = 2x$$

- What is the derivative of $g(x)$ at $x=5$?

$$\left. \frac{dg}{dx} \right|_{x=5} = 10$$

Example: Derivatives and Gradients

- What is the gradient of the function $g(x, y) = x^2y$
 - Recall: Gradient is a vector of partial derivatives with respect to each variable

$$\nabla g = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2xy \\ x^2 \end{bmatrix}$$

- What is the derivative of $g(x, y)$ at $x=0.5, y=0.5$?

$$\nabla g|_{x=0.5, y=0.5} = \begin{bmatrix} 2(0.5)(0.5) \\ (0.5^2) \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.25 \end{bmatrix}$$

Optimization Procedure: Gradient Ascent

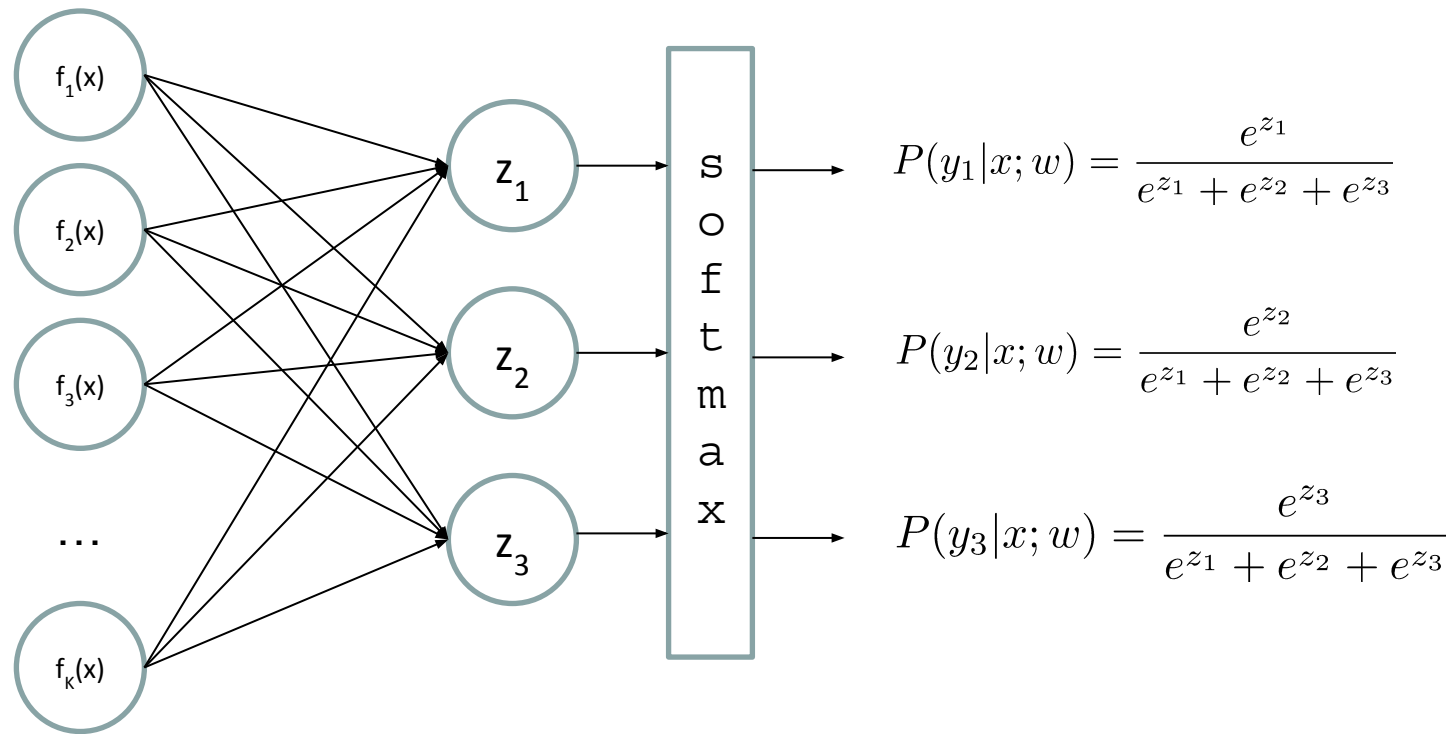
```
▪ init  $w$   
▪ for iter = 1, 2, ...
```

$$w \leftarrow w + \alpha * \nabla g(w)$$

- α : learning rate --- tweaking parameter that needs to be chosen carefully
- How? Try multiple choices
 - Crude rule of thumb: update changes w about 0.1 – 1 %

Multi-class Logistic Regression

- = special case of neural network



How about computing all the derivatives?

- Derivatives tables:

$$\frac{d}{dx}(a) = 0$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(au) = a \frac{du}{dx}$$

$$\frac{d}{dx}(u + v - w) = \frac{du}{dx} + \frac{dv}{dx} - \frac{dw}{dx}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{1}{v} \frac{du}{dx} - \frac{u}{v^2} \frac{dv}{dx}$$

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{1}{u}\right) = -\frac{1}{u^2} \frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{1}{u^n}\right) = -\frac{n}{u^{n+1}} \frac{du}{dx}$$

$$\frac{d}{dx}[f(u)] = \frac{d}{du}[f(u)] \frac{du}{dx}$$

$$\frac{d}{dx}[\ln u] = \frac{d}{dx}[\log_e u] = \frac{1}{u} \frac{du}{dx}$$

$$\frac{d}{dx}[\log_a u] = \log_a e \frac{1}{u} \frac{du}{dx}$$

$$\frac{d}{dx}e^u = e^u \frac{du}{dx}$$

$$\frac{d}{dx}a^u = a^u \ln a \frac{du}{dx}$$

$$\frac{d}{dx}(u^v) = vu^{v-1} \frac{du}{dx} + \ln u \cdot u^v \frac{dv}{dx}$$

$$\frac{d}{dx} \sin u = \cos u \frac{du}{dx}$$

$$\frac{d}{dx} \cos u = -\sin u \frac{du}{dx}$$

$$\frac{d}{dx} \tan u = \sec^2 u \frac{du}{dx}$$

$$\frac{d}{dx} \cot u = -\csc^2 u \frac{du}{dx}$$

$$\frac{d}{dx} \sec u = \sec u \tan u \frac{du}{dx}$$

$$\frac{d}{dx} \csc u = -\csc u \cot u \frac{du}{dx}$$